

KOELN BONN, Germany

WMO: 105130

Lat: 50.8644N

Lon: 7.1575E

Elev: 100

StdP: 100.14

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.0	-5.8	-13.0	1.2	-4.8	-10.0	1.6	-3.7	11.7	7.5	10.2	7.5	2.1	100	0.526

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	31.1	19.8	28.9	18.9	26.9	18.2	21.0	28.1	20.1	26.4	19.2	25.0	3.7	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.9	13.9	23.2	18.0	13.1	22.4	17.1	12.3	21.4	61.2	28.1	58.1	26.2	55.0	25.1	24.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.0	7.9	7.0	DB	-11.7	34.8	3.5	1.5	-14.3	35.9	-16.3	36.8	-18.3	37.6	-20.8	38.7	(4)
(5)			WB	-12.2	23.0	3.4	0.9	-14.7	23.6	-16.7	24.1	-18.6	24.6	-21.1	25.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.8	2.8	3.6	6.6	10.5	13.8	17.2	19.4	18.5	15.1	11.4	6.9	3.9	(6)	
	DBStd	6.91	4.31	4.25	3.99	4.02	3.85	3.53	3.49	3.14	3.03	3.75	3.87	4.29	(7)	
	HDD10.0	901	224	181	118	42	9	0	0	0	1	29	107	191	(8)	
	HDD18.3	2882	482	414	364	236	148	63	29	37	104	216	343	446	(9)	
	CDD10.0	1210	1	1	12	57	127	216	291	262	155	72	13	3	(10)	
	CDD18.3	151	0	0	0	1	8	29	62	41	9	1	0	0	(11)	
	CDH23.3	1496	0	0	0	31	95	300	591	399	75	5	0	0	(12)	
	CDH26.7	462	0	0	0	2	19	86	211	129	15	0	0	0	(13)	
(14) Wind	WSAvg	3.3	3.8	3.6	3.6	3.2	3.2	3.0	2.9	2.8	2.9	3.2	3.4	3.6	(14)	
(15) Precipitation	PrecAvg	850	67	61	56	51	65	88	86	87	69	69	73	76	(15)	
	PrecMax	1007	131	148	107	96	124	182	188	191	164	166	131	155	(16)	
	PrecMin	706	9	13	9	0	26	29	39	31	17	32	1	21	(17)	
	PrecStd	94	33	30	28	24	29	34	36	46	38	30	28	29	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.8	15.2	20.8	26.8	29.1	32.0	34.2	33.2	29.0	23.9	17.1	14.0	(19)	
		MCWB	10.8	8.6	11.5	14.5	18.5	21.0	20.5	20.4	19.1	17.2	13.3	11.1	(20)	
	2%	DB	11.8	12.2	17.2	23.8	25.9	29.1	31.7	30.1	25.2	20.6	15.1	12.6	(21)	
		MCWB	9.4	8.6	10.3	14.0	16.6	19.0	20.0	19.4	18.0	15.7	12.3	10.3	(22)	
	5%	DB	10.0	10.9	15.0	20.9	23.8	26.9	29.0	27.5	23.0	18.3	13.9	11.2	(23)	
		MCWB	8.2	8.1	9.7	12.8	15.6	18.4	19.0	18.6	17.1	14.7	11.4	9.1	(24)	
	10%	DB	8.5	9.3	12.9	18.0	21.5	24.5	26.8	25.1	20.9	16.9	12.2	9.9	(25)	
		MCWB	6.9	7.2	8.9	11.6	14.6	17.4	18.5	17.7	15.9	13.9	10.2	8.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.3	10.5	13.0	15.8	19.9	22.0	22.2	21.9	20.2	17.3	14.2	11.9	(27)	
		MCDB	13.2	13.1	17.5	23.4	26.2	29.4	30.1	29.9	26.6	21.4	16.5	13.4	(28)	
	2%	WB	9.7	9.3	11.5	14.4	18.1	20.4	21.1	20.5	18.6	16.3	12.7	10.6	(29)	
		MCDB	11.3	11.4	15.6	21.7	23.8	26.7	28.2	27.5	24.0	19.6	14.7	12.3	(30)	
	5%	WB	8.2	8.4	10.4	13.2	16.5	19.3	20.2	19.5	17.6	15.3	11.6	9.4	(31)	
		MCDB	9.8	10.5	14.0	19.3	21.7	25.1	26.6	25.4	21.9	18.1	13.5	11.1	(32)	
	10%	WB	6.9	7.3	9.3	12.2	15.2	18.1	19.3	18.5	16.5	14.2	10.5	8.1	(33)	
		MCDB	8.2	9.3	12.7	17.5	20.1	23.1	25.1	23.8	20.0	16.5	12.2	9.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.0	6.1	8.5	10.6	10.3	10.5	10.6	10.5	9.5	7.9	5.5	4.9	(35)	
		MCDBR	6.2	7.9	12.5	15.0	14.3	14.9	14.9	15.1	13.0	9.9	6.3	5.6	(36)	
	5% WB	MCWBR	5.1	5.2	6.9	7.1	6.7	6.8	5.7	6.4	6.6	5.8	4.5	4.5	(37)	
		MCWBR	5.9	7.2	10.3	12.9	12.1	12.5	12.7	12.4	11.1	8.8	5.6	5.4	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.317	0.343	0.374	0.410	0.415	0.418	0.423	0.416	0.386	0.362	0.338	0.315	(40)	
		taud	2.420	2.371	2.299	2.235	2.255	2.267	2.251	2.296	2.361	2.422	2.419	2.402	(41)	
	Ebn at Noon	Ebn at Noon	710	782	819	829	841	841	828	814	799	750	672	647	(42)	
		Edh at Noon	61	83	106	126	129	129	129	117	99	78	61	54	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	0.75	1.43	2.56	4.09	4.86	5.31	5.03	4.31	3.18	1.86	0.87	0.57	(44)	
	RadStd	RadStd	0.12	0.27	0.39	0.47	0.45	0.49	0.61	0.41	0.39	0.18	0.12	0.08	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (70 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page